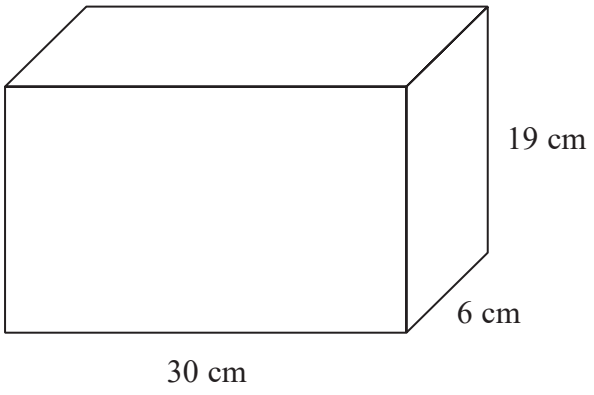
23 A container is in the shape of a cuboid.



The container is $\frac{2}{3}$ full of water.

A cup holds 275 m*l* of water.

What is the greatest number of cups that can be completely filled with water from the container?

Full volume = 30 × 6 × 19 = 3420 cm^3

1 mark
Multiply all three dimensions.

Each third = 3420 ÷ 3 = 1140 cm^3

1140	1140	1140

1 mark
Find two thirds of the full volume.

Water = 1140 × 2 = 2280 cm^3 = 2280 ml

2280 ÷ 275 = 8.2909... cups

1 mark
Divide by 275 ml per cup.

Only 8 cups can be completely filled.

1 mark
Round down to 8.

8

(Total for Question 23 is 4 marks)